

Deterministic geometric Savitzky-Golay dynamics estimator

Shared parameter update: reference -> estimated

mass [kg]	2.351597591	->	1.686295756	delta=-0.665302
ratio=0.71708517				
CoG x [m]	-0.002024708562	->	-0.02988379818	delta=-0.0278591
ratio=-				
CoG y [m]	-3.052657894e-05	->	-0.03543048732	delta=-0.0354 ratio=-
CoG z [m]	0.009509749599	->	0.05366855821	delta=+0.0441588
ratio=-				
rotor effectiveness 1	1	->	0.8692797992	delta=-0.13072 ratio=0.8692798
rotor effectiveness 2	1	->	0.7147961592	delta=-0.285204 ratio=0.71479616
rotor effectiveness 3	1	->	0.5815524887	delta=-0.418448 ratio=0.58155249
rotor effectiveness 4	1	->	0.6914601061	delta=-0.30854 ratio=0.69146011
command lag [s]	0	->	0.03283336053	delta=+0.0328334 ratio=-

Inertia matrix [kg m²]

reference:

[0.06500006148, -7.27899253e-07, 1.901508003e-05]
[-7.27899253e-07, 0.06495265634, 5.9167305e-08]
[1.901508003e-05, 5.9167305e-08, 0.1289921107]

estimated:

[0.008540177327, -0.001373977499, -1.049424687e-05]
[-0.001373977499, 0.00599206978, -0.003116697777]
[-1.049424687e-05, -0.003116697777, 0.01162592692]

Principal moments: reference -> estimated

principal moment 1 [kg m ²]	0.06495264516	->	0.00423613421	delta=-0.0607165
ratio=0.06521881				
principal moment 2 [kg m ²]	0.06500006701	->	0.008842952801	delta=-0.0561571
ratio=0.13604529				
principal moment 3 [kg m ²]	0.1289921163	->	0.01307908701	delta=-0.115913
ratio=0.10139447				

Optimization

joint residual-wrench accumulated squared loss: 306.855753546

Gaussian-prior cost: 535.867772688

selected strict-ZOH lag [s]: 0.032833360527

Bag failure_1_single

reconstruction position RMSE [m]: 69.3213368189

reconstruction position component RMSE [m]: [4.788761 3.687236 69.057366]

reconstruction orientation RMSE [deg]: 146.893067933

gyro consistency RMSE xyz [rad/s]: [21.328973 37.44924 20.615691]

specific-force consistency RMSE xyz [m/s²]: [24.220921 29.767387 21.712274]

inferred external force RMS xyz [N]: [6.814099 6.934208 6.392501]

inferred external torque RMS xyz [N m]: [0.662221 0.546709 0.672134]

legacy acceleration-residual loss diagnostic: 952.226695646

Recorded command timestamp intervals (data-derived)

Bag failure_1_single

rotor: median dt=0.00499773026 s (200.091 Hz), min=2.00271606e-05 s, max=0.0138893127 s

gimbal: median dt=0.00499868393 s (200.053 Hz), min=4.529953e-06 s, max=2.48282003 s

Parameter data objective

sample-summed uncentered squared dimensionless residual body wrench
force scale $F^* = 23.0612945139$ N
torque scale $\tau^* = 4.41819797713$ N m
residual mean is NOT subtracted before optimization

Selected optimizer diagnostics

status=0 success=False nfev=80 njev=16
message=The maximum number of function evaluations is exceeded.
reported optimality= $3.337655968493891e-06$
initial cost=842.7235388823646
final cost=842.7235262339176
cost reduction= $1.2648446954699466e-05$
coordinate step $L2=0.00047593948895730806$
final gradient inf-norm= $3.337655968493891e-06$
gradient tolerance satisfied=False
final Jacobian rank=13
final Jacobian $\text{cond}(\text{nonzero})=93.3238309291668$
max directional finite-difference Jacobian relative error= $1.1675691017274014e-07$